

Zhongqi Zhao

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INTERNSHIP

- The Chinese university of Hong Kong, Shenzhen** December 2025 - Present
Research Assistant Shenzhen, China

EDUCATION

- Donghua University** September 2018 - June 2022
BSc in Photoelectronic Information Science and Engineering Shanghai, China
- University of Copenhagen** September 2023 - June 2025
MSc in Physics Copenhagen, Denmark

PROJECTS

- Error mitigation for quantum chemistry algorithms on NISQ computers** September 2024 - June 2025
 - The project is about error mitigation techniques for quantum chemistry algorithms on NISQ computers. This work explores and extends a error mitigation scheme called Clifford Data Regression (CDR) to mitigate noise in quantum chemistry simulations using the Variational Quantum Eigensolver (VQE). Using the tiled Unitary Product State (tUPS) ansatz, I performed noisy simulation to investigate in detail the effect of various hyperparameters in CDR on the error mitigation quality. Building on these insights, two improvements to the CDR framework are proposed.
- A Clifford-Based Hamiltonian Engineering Approach via Reinforcement Learning** December 2025 - present
 - The project builds upon previous work in the research group on Hamiltonian engineering for quantum computational chemistry aimed to address the barren plateau problem. This approach increasing energy gradients at circuit initialization, thereby improving algorithm scalability and significantly reducing quantum resource requirements. The project has successfully demonstrated a complete 15-qubit VQE calculation on real quantum hardware. My role focuses on advancing the experimental implementation by incorporating complementary techniques, such as efficient measurement and error mitigation, to improve the accuracy and resource efficiency of VQE experiments.
- Anharmonic Vibrational Energy Levels with Variational Basis Encoding** August 2026 - present
 - Built a hybrid quantum-classical pipeline that combines Variational Basis Encoder (VBE) — which adaptively compresses each vibrational mode's physical basis into a small variational subspace optimized alongside the quantum circuit — with Variational Quantum Deflation (VQD) to compute ground and excited anharmonic vibrational energy levels of real polyatomic molecules using fewer qubits than direct encoding.

PUBLICATIONS

- Zhongqi Zhao**, Erik Rosendahl Kjellgren, Sonia Coriani, Jacob Kongsted, Stephan P. A. Sauer, Karl Michael Ziems. (2025). [Quantum error mitigation using energy sampling and extrapolation enhanced Clifford data regression](#). arXiv:2511.03556

SKILLS

- Programming Languages:** Python, C, C++
- Data Science & Machine Learning:** Scikit-learn, TensorFlow, PyTorch
- Specialized Area:** Qiskit, TensorCircuit, Pyscf
- Agent Applications:** Prompt Engineering, Tool Use, Skill Engineering

PROFESSIONAL MEMBERSHIPS

- Chinese Chemical Society** February 2026 - Present

CERTIFICATIONS

- Qiskit Global Summer School 2024 - Quantum Excellence** August 2024